Category: Cybersecurity

PS-013: Incident Timeline and Evidence Collection for Small Teams

- **Description:**
  - During a cybersecurity breach, solo defenders and small IT teams must manually reconstruct attack timelines from messy text logs scattered across Linux servers, web applications, and network appliances running with unaligned clocks and differing time zones. In this project, you will build a lightweight digital forensic tool. Your software will ingest raw host and application logs, compute cryptographic hashes upon collection to preserve evidence integrity, normalize timestamps into coordinated UTC, and assemble a searchable chronological incident timeline with an exportable summary report..
- **Challenge:**
  - Build a tool that collects host and application logs, normalises timestamps, preserves the original evidence and creates a searchable incident timeline. The system should help small teams reconstruct events quickly and consistently.
- **Key Requirements**
  - Process 2 different log formats.
  - Test against 3 synthetic incident scenarios.
  - Preserve original log records while creating normalised copies.
  - Generate evidence hashes for collected data.
  - Handle timezone differences and clock skew.
  - Identify and document parsing gaps.
  - Compare triage time against a manual baseline.
  - No live endpoint takeover, malware removal or forensic certification.
- **Expected Outcome**
  - Log ingestion and parsing tool
  - Timestamp normalisation module
  - Evidence preservation and hashing mechanism
  - Searchable incident timeline viewer
  - Three synthetic incident test cases
  - Incident-report export
  - Parsing-gap and triage-time analysis
  - Final technical demonstration and report
- **Deliverables***
  1. GitHub Collaboration: (**This step is crucial for allowing mentors to provide real-time oversight.**)
     - Each team must set up a private repository on GitHub.
     - Add our GitHub account (https://github.com/aiori-hackathon) as a collaborator.
     - Once the collaboration request has been accepted all the concerned users (all the teammates as well as aiori-hackathon account) transfer the ownership of the repository to our GitHub account. (https://github.com/aiori-hackathon)

- Ensure that your repository is named according to your team's name.

2. Each team is required to upload a PDF to their GitHub repository
   - Please follow the specified structure: Proposed-structure-hackathon.pdf
3. Prototype Demonstration with a Pseudocode Snippet to the mentor

- ## References
  - https://csrc.nist.gov/pubs/sp/800/86/final